



ENERGETIKA
ENERGETICS

Published from 2004
Ministry of Press and Information
of Azerbaijan Republic,
Registration number 3337, 07.03.2011

ISSN 1816-2126
Number 01, 2025
Section: English

ECOENERGETICS

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Comparative analysis of oil-contaminated soil remediation methods

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Abstract: The development of the oil industry, including accidents in oil production and processing, causes serious pollution of soil resources. Methods of remediation of oil-contaminated soils are being investigated to solve this problem. The main focus is on the method of recultivation, which allows restoration of natural ecosystem functions of oil-contaminated areas, re-formation of vegetation and normalization of soil structure.

Key words: Oil pollution, soil remediation, reclamation, ecological restoration.

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1. Introduction

Although the oil industry is important for the country's economy, it seriously pollutes the soil as a result of increasing leakages and accidents caused by the development of this field. Oil products change the composition of the soil and have a negative effect on plant and animal life. This problem poses a great threat to both agriculture and the overall ecosystem. Oil pollution usually occurs for two main reasons. Soils are polluted as a result of oil leakage into the soil from accidents that occur during oil production, transportation and processing, as well as technical malfunctions and leakages of equipment. When petroleum products are mixed with soil, the physical and chemical properties of the soil change. This inhibits the growth of plants, reduces the amount of microorganisms present in the soil and disturbs the stability of the ecosystem. As a result, agricultural productivity declines and ecological systems undergo a long-term recovery process [1-6].

2. Experimental detail

Various methods are used for soil restoration. The most commonly used methods are as follows.

- mechanical methods are based on the principle of removing the top layer of contaminated soil and adding clean soil instead. Although this method is used for immediate contamination removal, it may not be effective for deep contamination.

- chemical cleaning: it is possible to remove oil products from the soil by means of chemicals. However, this method sometimes disturbs the natural balance of the soil and may cause additional chemical waste.

- bioremediasiya və fitoremediasiya üsullarında mikroorqanizmlər və xüsusi bitkilərdən istifadə olunur.

They accelerate the natural regeneration of the soil by breaking down oil products. Bioremediation is the process by which contaminants in soil are broken down into harmless compounds by microorganisms. This method mainly uses the activity of bacteria, fungi and other microorganisms that are naturally present in the soil or artificially added. Bioremediation is carried out in two main ways:

- In situ bioremediation – (On-site treatment of contaminated soil). This method is cheaper and more environmentally friendly, but it can take a long time depending on the level of contamination.

- Ex situ bioremediation – (Cleaning contaminated soil by transferring it to another location). Although this method is faster, it can create transportation costs and environmental risks. Incorporating oil-degrading bacteria (such as *Pseudomonas putida* and *Alcanivorax borkumensis*) into the soil and adding nutrients such as nitrogen and phosphorus to the soil increases the efficiency of the bioremediation process.

- phytoremediation is the process of cleaning the soil from pollutants by means of plants. This method is environmentally safe and beneficial for soil structure. Examples of suitable plants for phytoremediation are sunflower (which has the ability to absorb heavy metals and oil pollutants in the soil), yellow mustard (accelerates the decomposition of oil waste), and poplar (cleans the soil by absorbing pollutants through its deep roots). Bioremediation and phytoremediation methods are environmentally efficient and safe solutions for the treatment of oil-contaminated

soils. Bioremediation cleans soil by breaking down pollutants using microorganisms, while phytoremediation uses plants to absorb and break down pollutants. The suitability of both methods depends on the level of contamination, soil conditions and ecosystem characteristics. A combination of these methods can make remediation of oil-contaminated soils more effective.

- in the thermal desorption method, oil-contaminated soil is cleaned by applying high temperature. Evaporation of oil and other organic pollutants under the influence of heat is the basis of this technique. As a result, pollutants are extracted from the soil, accumulated and then neutralized. When the soil is heated to a certain temperature by the thermal desorption method, oil and other organic pollutants turn into steam. After being collected by special systems, vaporized pollutants are neutralized. Contaminated soil is loaded in a special oven or thermal desorption device. A high temperature is applied to the soil. The temperature range is between 200 - 600°C, depending on the nature of the contaminants in the soil. Light organic materials vaporize at temperatures between 200 and 400°C, while heavier materials vaporize at higher temperatures.

Reclamation is the most widely used method for soil restoration. This method is also possible to bring the soil closer to its natural state. At this time, the soil structure is restored, local vegetation is restored and microbial activity is increased. Reclamation is beneficial both ecologically and economically. During the application of this method, the soil itself restores and creates conditions for rebuilding the local ecosystem. This method requires less investment than traditional mechanical and chemical methods. In the long-term perspective, protection of the ecological balance of the soil is ensured. When applying the reclamation method, first, the condition of the contaminated area is assessed. The chemical and physical properties of the soil, the level of pollution and the condition of the vegetation cover are determined. This assessment indicates which methods will be applied and how the reclamation process will be carried out. The use of local plant species is very important in the local reclamation process. These plants adapt to the soil and spread rapidly in the contaminated area and help to restore the soil structure. For example, species whose roots penetrate deep and grow quickly prevent soil erosion. Bioremediation measures are taken to increase the microbiological activity of the soil. As the number and variety of microorganisms increases, the decomposition of petroleum products and the natural recovery process are accelerated. Thus, for the success of the recultivation process, monitoring should be done regularly. The chemical, physical and biological parameters of the soil should be monitored, and additional measures should be taken if necessary. This can ensure the continuity of the recovery process and allow any problems to be eliminated at an early stage.

It is possible to assess the impact of mechanical, chemical and biological reclamation methods on the ecological restoration process in oil-contaminated soils. That is, an integrated approach can be more effective than single methods. Such research can be conducted in contaminated land areas near oil production areas. Samples were taken at different depths (0–30 cm and 30–60 cm), and the initial pollution level (oil-containing organic substances and hydrocarbon concentration) should be determined in laboratory conditions. The study should consist of three stages: The first stage should be the mechanical preparation stage. Excavation, mixing and aeration of the contaminated soil surface should be ensured. The second stage should be the stage of chemical processing. Decomposition of oil components by applying oxidizing agents should be applied in selected regions. Dosage and duration of administration should be determined based on prior pilot studies. The last stage is the stage of biological reclamation.

After mechanical and chemical measures, specific oil-degrading microorganisms and local plant species selected for phytoremediation (such as rhizomes and roses) should be added to the soil. The effect of the applied methods should be monitored within a period of 3 months. Every month, the data obtained from the soil samples should be evaluated based on the residual oil concentration, microbiological activity and plant growth indicators. As a result of the application of mechanical and chemical measures in the initial stages of the research, the concentration of oil-containing organic substances in the soil can be reduced by 35-40% on average.

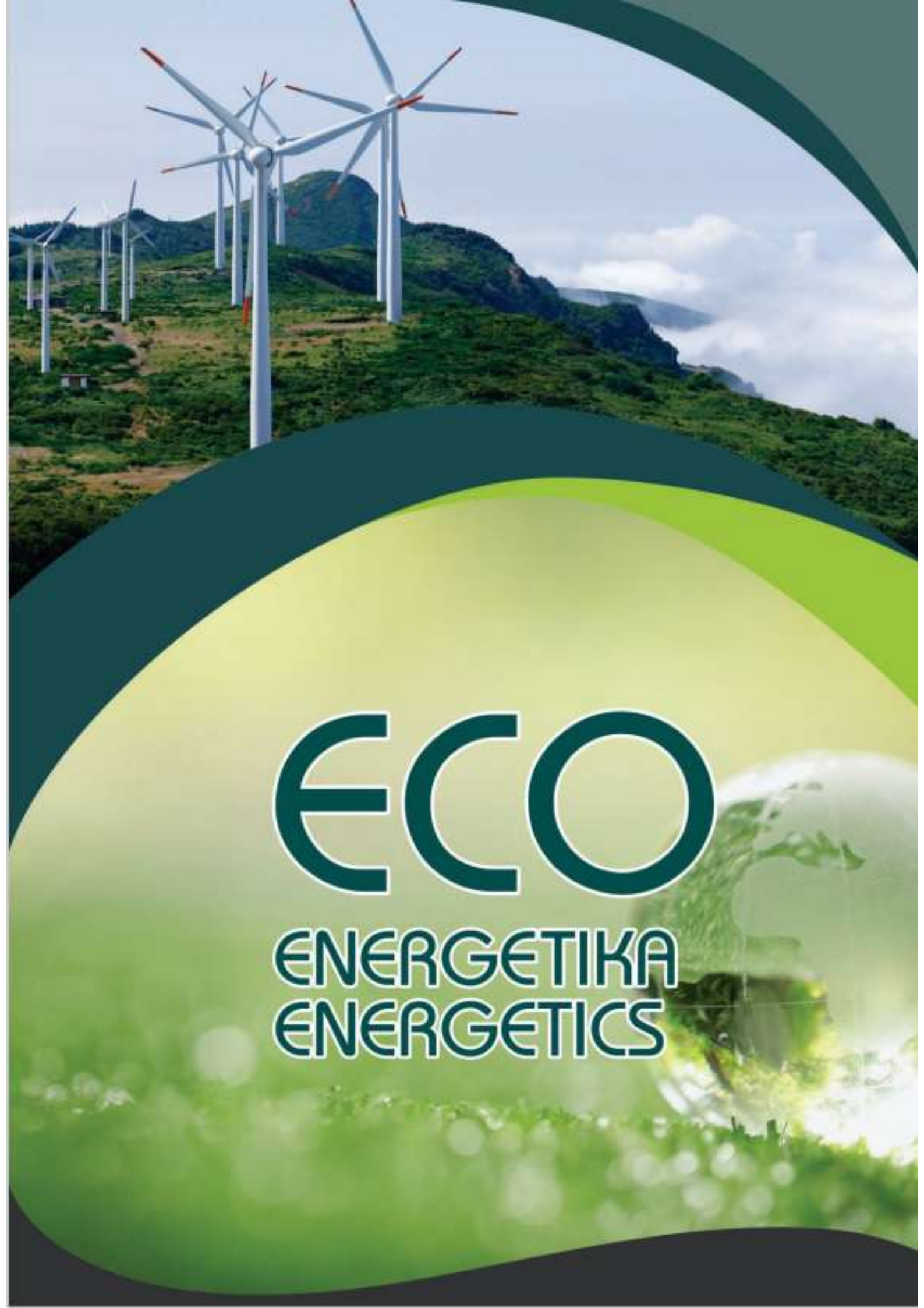
Later, as a result of the biological reclamation phase, the population of oil-degrading bacteria and fungi in the soil can increase by up to 50%. The root system of the plants planted in the recultivated areas is denser and more developed, and up to 60% of the surface vegetation can be restored. As a result of the application of the integrated method, the nutrient and microelement balance of the soil can be significantly improved. That is, compared to the use of only mechanical and chemical methods, the addition of a biological approach can enable quick and effective soil restoration. The results obtained in the study show that the integration of mechanical, chemical and biological methods can allow the restoration of contaminated soil in a short time and significantly. The large-scale application of this approach creates important prospects for the restoration of ecosystems and the increase of agricultural productivity. Compared to other methods, reclamation methods are less expensive and have minimal environmental side effects.

3. Conclusion

Remediation of oil-contaminated soils, restoration of ecosystem health, and sustainability of agriculture are important issues. Although there are various methods for soil restoration, the most superior and sustainable method is recultivation. Reclamation normalizes the chemical, physical and biological properties of the soil by supporting natural processes. With the use of local plant species and the support of microorganisms, this method ensures both environmental safety and economic benefits. In addition, in cases of high pollution, it is important to take additional measures and conduct continuous monitoring. In future studies, the effectiveness of the reclamation method in different climatic conditions and soil types should be investigated more widely.

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The background of the entire image is a photograph of a wind farm. Several white wind turbines with three blades each are positioned on a green, grassy hill. The sky is blue with some white clouds. The image is framed by a dark green curved border at the top and bottom. A large, light green semi-circular shape is overlaid on the bottom half of the image, containing the text.

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